



EXAMINATIONS SEPTEMBER /OCTOBER 2020
SUPPLEMENTARY SEMESTER / GRADE IMPROVEMENT/
RE -REGISTERED CANDIDATES

Program	: B.E. : Computer Science and Engg	Semester	: V
Course Name	: Database Systems	Max. Marks	: 100
Course Code	: CS52	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT- I

- What is DBMS? What are the advantages of using DBMS? CO1 (08)
 - Construct an E-R diagram for an airline. The database must keep track of customers and their reservations, flights and their status, seat assignments on individual flights and the schedule and routing of future flights. Specify the key attributes of each entity type, the participation and the cardinality ratio. CO1 (08)
 - Explain any 4 different types of user friendly interfaces. CO1 (04)
- With a neat diagram, explain the component modules of a DBMS CO1 (08)
 - A University database contains information about Instructors (identified by a Social security number) and Courses(identified by a course ID). Each of the following situations contains the relationship set between the teacher and the student. Design an ER diagram for each situation(assuming that no further constraints hold)
 - Instructors can teach the same course over several semesters and each offering must be recorded.
 - Each Instructor teaches exactly one Course.
 - Each Instructor teaches at least one Course and some Instructors may teach multiple courses.
 - Each Instructor teaches at least one Course and some Instructors may teach multiple courses.
 - Explain the concept of Ternary relationships with example. CO1 (04)

UNIT- II

- Define union compatibility of relations. Explain the set theoretical operations in relational algebra with example. CO2 (06)
 - Assume that you have two relations R & S where R is parent relation and S is child relation. R has primary key values where S references them. In a situation you are about to delete some tuples in relation R which causes inconsistency in relation S. As a designer what solution you might suggest to avoid inconsistencies in relation S? CO2 (04)
 - Consider the following schema: CO2 (10)
 - Suppliers(sid, sname , address)
 - Parts(pid, pname, color)
 - Catalog(sid, pid, cost)

The key fields are underlined. Therefore sid is the key for Suppliers, pid is the key for Parts, and sid and pid together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in relational algebra

1. Find the names of suppliers who supply some red part.
 2. Find the sids of suppliers who supply some red or green part.
 3. Find the sids of suppliers who supply some red part or are at Mysore.
 4. Find the sids of suppliers who supply some red part and some green part.
 5. Find the sids of suppliers who supply every part.
4. a) Discuss the division operation in relational algebra with an example. CO2 (04)
 b) Consider the following two relations. Show the resultant relations after applying left outer join, right outer join and full outer join. CO2 (07)

R			S	
ID	Name		ID	Address
1	Abhi		1	Delhi
2	Adam		2	Mumbai
3	Alex		3	Chennai
4	Anu		7	Noida
5	Ashih		8	Panipat

- c) Explain the rename operation in relational algebra with an example. CO2 (04)
 d) Discuss the aggregate functions of relational algebra with its syntax by considering relational schema that has at least one numerical attribute. Display the results after you apply each operation. CO2 (05)

UNIT- III

5. a) Explain with examples, the various JOIN operations possible in SQL. CO3 (08)
 b) Write a note on impedance mismatch. CO3 (06)
 c) Consider the schema given below: CO3 (06)
 EMP(Ssn, fname, lname, bdate, address, sex, salary, superssn, dno)
 DEPT(dnumber, dname, mgrssn, mgrstartdate)
 PROJECT(pnumber, pname, ploc, dnum)
 WORKS_ON(essn, pno, hours)
 DEPENDENT(Dname, Essn)
 Write the SQL statements for the following queries:
 i. List the names of all department managers who have no dependents
 ii. Increase the salary of employees of research department by 5%
 iii. Delete employees of mechanical department.
6. a) Consider the schema given below for a aircraft database. CO3 (08)
 Aircraft (aircraft ID, aircraft name, cruising range)
 Certified (emp ID, aircraft ID)
 Employee (Emp ID, ename, salary)
 Write the SQL statements for the following queries:
 i. Find the employee ID's of employee who make the highest salary.
 ii. Find the employees who are certified for the largest no of aircrafts.
 iii. Arrange the flight no with respect to the ascending order of distance.
 iv. Find the name of pilots who can operate flights with a range greater than 3000 miles but are not certified on any Boeing aircraft.
- b) Differentiate between two approaches of view implementation. CO3 (06)
 c) What is Trigger? What is its need? How to create assertion for a relation? Explain with examples. CO3 (06)

UNIT- IV

7. a) List and explain the informal guidelines for a relational schema design. CO4 (08)
b) Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies $F = \{ \{A, B\} \rightarrow C, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$. What is the key for R? Decompose R into 2NF and then into 3NF relations. CO4 (08)
c) Prove the following Inference Rules for functional dependencies CO4 (04)
i. $\{X \rightarrow Y, XY \rightarrow Z\} \vdash \{X \rightarrow Z\}$
ii. $\{X \rightarrow Y, Z \rightarrow W\} \vdash \{XZ \rightarrow YW\}$
8. a) Explain normal forms based on primary keys and super keys and functional dependencies with examples. CO4 (08)
b) Given the set of FD's, $F: \{ A \rightarrow BC, B \rightarrow CE, A \rightarrow E, AC \rightarrow H, D \rightarrow B \}$. Find the minimal cover of F. CO4 (06)
c) Write the algorithm for testing of non-additive join property. CO4 (06)

UNIT- V

9. a) What is meant by the concurrent execution of database transactions in a multiuser system? Discuss why concurrency control is needed, and give informal examples. CO5 (07)
b) Discuss the actions taken by the read_item and write_item operations on a database. CO5 (06)
c) What is the system log used for? What are the typical kinds of records in a system log? What are transaction commit points, and why are they important? CO5 (07)
10. a) Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules. CO5 (10)
a. $r_1(X); r_3(X); w_1(X); r_2(X); w_3(X);$
b. $r_1(X); r_3(X); w_3(X); w_1(X); r_2(X);$
c. $r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$
d. $r_3(X); r_2(X); r_1(X); w_3(X); w_1(X)$
b) What is the two-phase locking protocol? How does it guarantee serializability? CO5 (06)
c) What is a timestamp? How does the system generate timestamps? CO5 (04)
